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Light-emitting element and display panel including a light guide structure to improve optical quality

Abstract

A light-emitting element including an epitaxial structure and a light guide structure is provided. The epitaxial structure has a first surface and a second surface opposite to each other and includes an active layer. The light guide structure is disposed on the first surface of the epitaxial structure and includes a first light reflection layer and a filter layer covering on an upper surface of the first light reflection layer. The first light reflection layer completely covers the first surface. A display panel is also provided.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application Ser. No. 110108846, filed on Mar. 12, 2021. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to a semiconductor device, and particularly relates to a light-emitting element and a display panel.

Description of Related Art

(3) Color presentation of a display panel may directly affect viewer's perception, so that the display panel must have a higher chromaticity specification to achieve a better visual experience. Furthermore, color purity of light emitted by a light-emitting element in the display panel may have a considerable impact on optical quality thereof. For example, when the color purity of the emitted light is higher, the display panel may have better optical quality, so that how to improve the color purity of the light emitted by the light-emitting element to make the display panel to have better optical quality is a very important issue in the field.

SUMMARY

(4) The disclosure is directed to a light-emitting element and a display panel, which are adapted to improve color purity of light emitted by the light-emitting element, so that the display panel has better optical quality.

(5) The disclosure provides a light-emitting element including an epitaxial structure and a light guide structure. The epitaxial structure has a first surface and a second surface opposite to each other and includes an active layer. The light guide structure is disposed on the first surface of the epitaxial structure and includes a sequentially stacked first light reflection layer and a filter layer covering an upper surface of the first light reflection layer. The first light reflection layer completely covers the first surface.

(6) In an embodiment of the disclosure, the filter layer partially covers the upper surface of the first light reflection layer.

(7) In an embodiment of the disclosure, the first light reflection layer includes a first region covered by the filter layer and a second region exposed outside the filter layer, and a ratio of an area of the first region to an area of the upper surface is greater than or equal to 20%.

(8) In an embodiment of the disclosure, the light-emitting element further includes at least one electrode electrically connected to the epitaxial structure, and the at least one electrode is adjacent to the filter layer and disposed on the upper surface.

(9) In an embodiment of the disclosure, the filter layer completely covers the upper surface of the first light reflection layer.

(10) In an embodiment of the disclosure, the light-emitting element further includes a second light reflection layer, and the second light reflection layer covers the second surface.

(11) In an embodiment of the disclosure, the epitaxial structure further has a side surface connecting the first surface and the second surface, and the second light reflection layer further covers the side surface.

(12) In an embodiment of the disclosure, the first light reflection layer extends beyond the side

surface of the epitaxial structure, and the second light reflection layer is connected to the first light reflection layer.

(13) In an embodiment of the disclosure, the light-emitting element further includes a first protective layer, and the light guide structure is located between the first protective layer and the epitaxial structure, or the first protective layer is located between the first light reflection layer and the epitaxial structure.

(14) In an embodiment of the disclosure, the light-emitting element further includes a second protective layer, and the second protective layer is disposed at both sides of the filter layer and covers the first light reflection layer.

(15) In an embodiment of the disclosure, the light-emitting element further includes a plurality of electrodes electrically connected to the epitaxial structure. The plurality of electrodes are disposed on the second surface of the epitaxial structure; or the plurality of electrodes are respectively disposed on the first surface and the second surface of the epitaxial structure.

(16) In an embodiment of the disclosure, a thickness of the filter layer is positively correlated with an equivalent refractivity of the light guide structure.

(17) In an embodiment of the disclosure, the first light reflection layer is doped with a P-type material or an N-type material.

(18) In an embodiment of the disclosure, the first light reflection layer and the filter layer include an aluminum material, and aluminum content of the first light reflection layer is less than aluminum content of the filter layer.

(19) The disclosure provides a display panel including a substrate and a plurality of light-emitting elements. The light-emitting elements are arranged on the substrate. The light-emitting elements have different color light and are electrically connected to the substrate. Each of the light-emitting elements includes an epitaxial structure and a light guide structure. The epitaxial structure has a first surface and a second surface opposite to each other and includes an active layer. The light guide structure is disposed on the first surface of the epitaxial structure and includes a first light reflection layer and a filter layer covering an upper surface of the first light reflection layer. The first light reflection layer completely covers the first surface.

(20) In an embodiment of the disclosure, an area of an orthographic projection of the filter layer of each of the light-emitting elements on the substrate is less than or equal to an area of an orthographic projection of the corresponding epitaxial structure on the substrate.

(21) In an embodiment of the disclosure, the first light reflection layers of the light-emitting elements are connected to each other.

(22) In an embodiment of the disclosure, the protective layer is located between the first light reflection layers and the epitaxial structures of the light-emitting elements.

(23) In an embodiment of the disclosure, a gap exists between any two adjacent filter layers, and the first light reflection layers are exposed to the outside by the gap.

(24) In an embodiment of the disclosure, the light-emitting elements respectively have first color light, second color light, and third color light. Each of the light-emitting elements includes a first filter layer allowing the first color light to pass through, a second filter layer allowing the second color light to pass through, and a third filter layer allowing the third color light to pass through. The first filter layer, the second filter layer and the third filter layer are sequentially stacked on the first light reflection layer.

(25) In an embodiment of the disclosure, the epitaxial structure further has a side surface connecting the first surface and the second surface. Each of the light-emitting elements further includes a second light reflection layer, and the second light reflection layer covers the second surface and the side surface.

(26) Based on the above description, in the light-emitting element of the disclosure, through combination of the first light reflection layer and the filter layer, only light of a specific wavelength may be transmitted out of the light guide structure. In this way, the wavelength (color) of the

emitted light may be controlled within a small range, thereby improving the color purity of the light emitted by the light-emitting elements, so that the display panel including the light-emitting elements has better optical quality.

(27) To make the aforementioned more comprehensible, several embodiments accompanied with drawings are described in detail as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.

(2) FIG. 1-FIG. 12 are schematic partial cross-sectional views of light-emitting elements according to some embodiments of the disclosure.

(3) FIG. 13A, FIG. 14A, FIG. 15A, and FIG. 16A are partial schematic cross-sectional views of display panels according to some embodiments of the disclosure.

(4) FIG. 13B, FIG. 14B, FIG. 15B, and FIG. 16B are respectively partial enlarged view of a region A of FIG. 13A, a region B of FIG. 14A, a region C of FIG. 15A, and a region D of FIG. 16A.

DESCRIPTION OF THE EMBODIMENTS

(5) Detailed features and advantages of the disclosure are described in detail in following embodiments, and the content thereof is sufficient to enable anyone familiar with the related art to understand the technical content of the disclosure and implement the same accordingly, and according to the content disclosed in this specification, a scope of patent application and the drawings, anyone familiar with the related art may easily understand the purpose and advantages of the disclosure. The following examples further illustrate the viewpoints of the disclosure in detail, but the scope of the disclosure is not limited by any viewpoint.

(6) FIG. 1-FIG. 12 are schematic partial cross-sectional views of light-emitting elements according to some embodiments of the disclosure. Referring to FIG. 1, in the embodiment, a light-emitting element **100** includes an epitaxial structure **110** and a light guide structure **120**, wherein the epitaxial structure **110** has a first surface **110a** and a second surface **110b** opposite to each other, and the light guide structure **120** is disposed on the first surface **110a** of the epitaxial structure **110**. Furthermore, the epitaxial structure **110** may include a first-type semiconductor layer **112**, an active layer **114**, and a second-type semiconductor layer **116** stacked in sequence, and the light guide structure **120** includes a first light reflection layer **122** and a filter layer **124**.

(7) In some embodiments, the first-type semiconductor layer **112** of the epitaxial structure **110** is, for example, a P-type semiconductor layer, such as P-GaN, and the active layer **114** is, for example, a multiple quantum well (MQW) structure, while the second-type semiconductor layer **116** is, for example, an N-type semiconductor layer, such as N—GaN, but the disclosure is not limited thereto, and may be adjusted according to actual design requirements.

(8) In the embodiment, the first light reflection layer **122** of the light guide structure **120** completely covers the first surface **110a**, and the filter layer **124** covers an upper surface **122a** of the first light reflection layer **122**. Therefore, through the combination of the first light reflection layer **122** and the filter layer **124**, the first light reflection layer **122** may form a forbidden band, and the filter layer **124** may be used to change an equivalent refractivity of the covered first light reflection layer **122**, so as to form a constructive interference section corresponding to a specific transmission spectrum in the forbidden band to only allow light with a wavelength within such section to pass through. Therefore, by covering the upper surface **122a** of the first light reflection layer **122** with the filter layer **124**, when the light projected to the first light reflection layer **122**, the

light is shielded by the forbidden band. If a wavelength of the light is matched with the equivalent refractivity of the filter layer **124** (and the first light reflection layer **122**), it represents that the light is constructively interfered with an incident interface, so that only light of a specific wavelength may be transmitted through the light guide structure **120**. In this way, the wavelength (color) of the emitted light may be controlled within a small range, thereby improving the color purity of the emitted light of the light-emitting element **100**. In here, the light of the light-emitting element **100** may be emitted from the active layer **114** toward the light guide structure **120**, and the upper surface **122a** of the first light reflection layer **122** may be opposite to a lower surface **122b**, wherein the lower surface **122b** directly contacts the first surface **110a** of the epitaxial structure **110**.

(9) In some embodiments, the first light reflection layer **122** and the filter layer **124** may be a multilayer structure disposed on the epitaxial structure **110**. For example, the filter layer **124** may include one or more layers of refractivity matching layer **126**, and changes in configuration of the material of each layer of the filter layer **124** and the refractivity matching layer **126** (such as but not limited to changes in material, thickness, sequence, number, etc.) may determine a wavelength of light that is allowed to pass through the filter layer **124**. From an optical perspective, an overlapping portion of the first light reflection layer **122** and the filter layer **124** in a stacking direction of the epitaxial structure **110** may be regarded as a band pass filter (BPF), and a portion of the first light reflection layer **122** that is not overlapped with the filter layer **124** in the stacking direction of the epitaxial structure **110** may be regarded as a distributed Bragg reflector (DBR) (which is not shown in the embodiment), but the disclosure is not limited thereto.

(10) In some embodiments, the thickness, material, and arrangement of each layer in the multilayer structure of the first light reflection layer **122** and the filter layer **124** may affect an overall equivalent refractivity and a transmissible wavelength. For example, the thicker a single layer in the multilayer structure of the first light reflection layer **122** and the filter layer **124** is and the denser the multilayer arrangement is (the smaller a distance between the layers is), the larger the overall equivalent refractivity is, and the larger the transmissible wavelength is, but the disclosure is not limited thereto.

(11) In some embodiments, the material of the first light reflection layer **122**, the filter layer **124** and the refractivity matching layer **126** include SiO_2 , TiO_2 , Ta_2O_5 , Ta_xO_y , MgF_2 , SiN_x , SiON or a combination thereof, but the disclosure is not limited thereto.

(12) In some embodiments, the first light reflection layer **122** and the filter layer **124** are doped with a P-type material or an N-type material, the first light reflection layer **122** and the filter layer **124** include aluminum materials, and aluminum content of the first light reflection layer **122** is less than aluminum content of the filter layer **124**. For example, the material of the first light reflection layer **122** and the material of the filter layer **124** include $\text{Al}_x\text{Ga}_{1-x}\text{N}$, $\text{Al}_x\text{Ga}_{1-x}\text{InN}$, $\text{Al}_x\text{Ga}_{1-x}\text{InP}$, $\text{Al}_x\text{Ga}_{1-x}\text{As}$, or a combination thereof. The smaller the value of x is, the lower the aluminum content is, and the higher the refractivity is. Therefore, the aluminum content of the first light reflection layer **122** may be less than the aluminum content of the filter layer **124**, so that the refractivity of the first light reflection layer **122** may be greater than the refractivity of the filter layer **124**, but the disclosure is not limited thereto.

(13) In some embodiments, the material of the first light reflection layer **122** and the material of the filter layer **124** may be the same, but the disclosure is not limited thereto, and the material of the first light reflection layer **122** and the material of the filter layer **124** may also be different.

(14) In some embodiments, the first light reflection layer **122** includes a region R covered by the filter layer **124**, wherein a ratio of an area of the region R to an area of the upper surface **122a** is greater than or equal to 20%, so as to reduce production difficulty of the film layers. For example, the filter layer **124** of the embodiment may completely cover the upper surface **122a** of the first light reflection layer **122**, i.e., the ratio of the area of the region R to the area of the upper surface **122a** is 100%, but the disclosure is not limited thereto, and other embodiments may have different

coverage patterns.

(15) In some embodiments, the refractivity matching layer **126** covers and contacts the upper surface **122a** of the first light reflection layer **122**, so that the equivalent refractivity of the light guide structure **120** may be changed by adjusting the refractivity matching layer **126** in the filter layer **124**, but the disclosure is not limited thereto.

(16) In some embodiments, the thickness of the refractivity matching layer **126** is positively related with the equivalent refractivity of the light guide structure **120**, and an expression thereof may be $d = \lambda / 2n$, wherein d is the thickness, λ is the wavelength, and n is the refractivity. For example, when the thickness of the refractivity matching layer **126** becomes thicker, the equivalent refractivity becomes higher and a critical refraction angle becomes smaller, so that a light shape is narrowed and a light output range becomes smaller, and vice versa. When the thickness of the refractivity matching layer **126** becomes thinner, the equivalent refractivity becomes lower and the critical refraction angle becomes larger, so that the light shape is broadened, and the light output range becomes larger. Therefore, by adjusting the thickness of the refractivity matching layer **126**, a total internal reflection angle and the transmission angle of the filter layer **126** are controlled within a required range, but the disclosure is not limited thereto.

(17) Accordingly, in the light-emitting element **100** of the embodiment, the equivalent refractivity of the overall structure is adjusted through the design of the first light reflection layer **122** and the filter layer **124** in the light guide structure **120**, so that the light guide structure **120** only allows photons of a specific wavelength to pass through in each wavelength range. Namely, only light of a specific wavelength may be transmitted out of the light guide structure **120**. In this way, the wavelength (color) of the emitted light may be controlled within a small range, which improves the color purity of the light emitted by the light-emitting element **100**.

(18) It should be noted that reference numbers of the components and a part of contents of the aforementioned embodiment are also used in the following embodiment, wherein the same reference numbers denote the same or like components, and descriptions of the same technical contents are omitted. The aforementioned embodiment may be referred for descriptions of the omitted parts, and detailed descriptions thereof are not repeated in the following embodiment.

(19) Referring to FIG. 2, compared with the light-emitting element **100**, a filter layer **224** of a light guide structure **220** of a light-emitting element **200** of the embodiment partially covers the upper surface **122a** of the first light reflection layer **122**. In other words, the filter layer **224** may be retracted within the first light reflection layer **122**, so that the light shape may be controlled within a desired light output range, and probability of light divergence may be reduced. For example, an angle at which light L1 enters the first light reflection layer **122** is θ_1 , and a light output point thereof is located between side edges **224e** of the filter layer **224**, so that the light L1 may be transmitted out, and an angle at which light L2 enters the first light reflection layer **122** is θ_2 , and a light output point thereof is coincided with the side edge **224e** of the filter layer **224**. Namely, when the angle of the light incident to the first light reflection layer **122** is greater than θ_2 , the light is subjected to a total internal reflection and cannot be transmitted out, but the disclosure is not limited thereto.

(20) In the embodiment, the first light reflection layer **122** may include a region R1 covered by the filter layer **224** and a second region R2 exposed outside the filter layer **224**, wherein the first region R1 is sandwiched between two adjacent first regions R1. Furthermore, a ratio of an area of the first region R1 to an area of the upper surface **122a** may be greater than or equal to 20% and less than 100%, but the disclosure is not limited thereto.

(21) Referring to FIG. 3, compared with the light-emitting element **200**, a light-emitting element **300** of the embodiment further includes a second light reflection layer **330** covering the second surface **110b**, wherein the second light reflection layer **330** may have a function of a distributed Bragg reflector, through the design of the distributed Bragg reflector, the light emitted downward from the active layer **114** may be reflected back for recycling, and the recycled light may be

emitted from the top to increase a light output of the light-emitting element **300**. The second light reflection layer **330** may be similar to the first light reflection layer **122**, which is omitted.

(22) In the embodiment, the second light reflection layer **330** may completely cover the second surface **110b** of the epitaxial structure **110** and does not extend beyond a side surface **110s** of the epitaxial structure **110**, wherein the side surface **110s** is a surface of the epitaxial structure **110** connecting the first surface **110a** and the second surface **110b**, but the disclosure is not limited thereto. In some embodiments, the second light reflection layer may be retracted within the side surface **110s** of the epitaxial structure **110**, or the second light reflection layer may extend beyond the side surface **110s** of the epitaxial structure **110**.

(23) Referring to FIG. 4, compared with the light-emitting element **300**, a second light reflection layer **430** of a light-emitting element **400** of the embodiment further covers a side surface **410s** of an epitaxial structure **410** to protect the epitaxial structure **410**, so that a part of light may be reflected in the epitaxial structure **410** and then emitted out, and when the light-emitting elements are subsequently applied to the display panel in an array manner, a phenomenon of lateral light mixing may be prevented, wherein the side surface **410s** is a surface of the epitaxial structure **410** connecting a first surface **410a** and a second surface **410b**. In here, a cross-sectional profile of the epitaxial structure **410** may be a trapezoidal profile. Therefore, a side surface of a first-type semiconductor layer **412**, a side surface of an active layer **414**, and a side surface of a second-type semiconductor layer **416** in the epitaxial structure **410** may be obliquely extended and connected, but the disclosure is not limited thereto. In an embodiment that is not illustrated, a cross-sectional profile of an epitaxial structure may also be a rectangular profile as shown in FIG. 1 to FIG. 3. It should be noted that the cross-sectional profiles of the epitaxial structures in various embodiments of the disclosure may all be trapezoidal profiles or rectangular profiles.

(24) In the embodiment, a first light reflection layer **422** of a light guide structure **420** extends beyond the side surface **410s** of the epitaxial structure **410**, and the second light reflection layer **430** is connected to the first light reflection layer **422** so that the first light reflection layer **422** and the second light reflection layer **430** surround the epitaxial structure **410**, but the disclosure is not limited thereto.

(25) In the embodiment, the first light reflection layer **422** may include a first region **R11** covered by the filter layer **224** and a second region **R21** exposed outside the filter layer **224**, wherein the filter layer **224** may partially cover an upper surface **422a** of the first light reflection layer **422**. In other words, a ratio of an area of the first region **R11** to an area of the upper surface **422a** is less than 100%, but the disclosure is not limited thereto.

(26) Referring to FIG. 5, compared to the light-emitting element **100**, a first light reflection layer **522** of a light guide structure **520** of a light-emitting element **500** of the embodiment extends beyond the side surface **110s** of the epitaxial structure **110**. In other words, the filter layer **124** may partially cover an upper surface **522a** of the first light reflection layer **522**. In addition, the light-emitting element **500** may further include a first protective layer **540** disposed on the light guide structure **520**, wherein the light guide structure **520** may be located between the first protective layer **540** and the epitaxial structure **110**, but the disclosure is not limited thereto. In other embodiments, the first protective layer may have other configurations. A material of the first protective layer **540** may include resin or glass.

(27) In some embodiments, the light guide structure **520** is pre-formed on the first protective layer **540**, and then the light guide structure **520** and the first protective layer **540** are soft-bonded to the epitaxial structure **110** to reduce a chance of damaging the light-emitting element caused by a stress formed when the light guide structure **520** is formed on the epitaxial structure **110**, wherein the soft bonding is performed by using low-refractive light-transmitting media such as epoxy resin, air (or vacuum), but the disclosure is not limited thereto.

(28) In some embodiments, the light guide structure **520** is first formed on the epitaxial structure **110**, and then the first protective layer **540** is bonded to the light guide structure **520**, but the

disclosure is not limited thereto.

(29) Referring to FIG. 6, compared to the light-emitting element **500**, a first protective layer **640** of a light-emitting element **600** of the embodiment is located between the first light reflection layer **522** and the epitaxial structure **110**. In other words, the first protective layer **640** may directly contact the second-type semiconductor layer **116** of the epitaxial structure **110**, but the disclosure is not limited thereto.

(30) In some embodiments, similar to the first protective layer **540**, the light guide structure **520** may be pre-formed on the first protective layer **640**, and then the light guide structure **520** and the first protective layer **640** are soft-bonded to the epitaxial structure **110** to reduce the chance of damaging the light-emitting element caused by a stress formed when the light guide structure **520** is formed on the epitaxial structure **110**, but the disclosure is not limited thereto.

(31) Referring to FIG. 7, compared to the light-emitting element **600**, a light-emitting element **700** of the embodiment further includes a second protective layer **750**. The second protective layer **750** is disposed at both sides of the filter layer **124** and covers the first light reflection layer **522** to further enhance protection on the light-emitting element **700**, but the disclosure is not limited thereto. A material of the second protective layer **750** may include resin or glass.

(32) In some embodiments, the second protective layer **750** surrounds the filter layer **124**, but the disclosure is not limited thereto.

(33) Referring to FIG. 8, compared to the light-emitting element **200**, a light-emitting element **800** of the embodiment further includes a plurality of electrodes (such as a first electrode **861** and a second electrode **862** in FIG. 8), wherein the first electrode **861** and the second electrode **862** are disposed on the side of the second surface **110b** of the epitaxial structure **110**, and are electrically connected to the epitaxial structure **110**. In other words, the first electrode **861** and the second electrode **862** may be horizontally arranged on the epitaxial structure **110**.

(34) In some embodiments, the first electrode **861** is a first-type electrode and is electrically connected to the first-type semiconductor layer **112**, and the second electrode **862** is a second-type electrode and is electrically connected to the second-type semiconductor layer **116**, but the disclosure is not limited thereto. A material of the first electrode **861** and the second electrode **862** is, for example, metal, but the disclosure is not limited thereto.

(35) In an embodiment that is not shown, the first electrode **861** and the second electrode **862** are disposed on the side of the first surface **110a** of the epitaxial structure **110**, wherein the first electrode **861** may penetrate through all of the first light reflection layer **122**, the second-type semiconductor layer **116**, and the active layer **114** to be in contact with and electrically connected to the first-type semiconductor layer **112**. The second electrode **862** is in contact with and electrically connected to the second-type semiconductor layer **116**, but the disclosure is not limited thereto.

(36) Referring to FIG. 9, compared to the light-emitting element **800**, the second light reflection layer **330** of a light-emitting element **900** of the embodiment is doped with a P-type material or an N-type material, so that the first electrode **961** and the second electrode **962** may directly make ohmic contact with the second light reflection layer **330**. In addition, the light-emitting element **900** further includes a passivation layer **970** to reduce a chance of current leakage between the first electrode **961** and the second electrode **962**, but the disclosure is not limited thereto.

(37) Referring to FIG. 10, compared to the light-emitting element **900**, a light-emitting element **1000** of the embodiment further includes a second light reflection layer **1030** in a stair-step shape, and a first electrode **1061** and a second electrode **1062** may also have a stair-step shape corresponding to the second light reflection layer **1030**, but the disclosure is not limited thereto.

(38) Referring to FIG. 11, compared to the light-emitting element **1000**, a plurality of electrodes (a first electrode **1161** and two second electrodes **1162** in FIG. 11) of a light-emitting element **1100** of the embodiment are respectively disposed on the first surface **110a** and the second surface **110b** of the epitaxial structure **110**, and are electrically connected to the epitaxial structure **110**. In other

words, the first electrode **1161** and the two second electrodes **1162** are vertically arranged on the epitaxial structure **110**. Furthermore, in the embodiment, the two second electrodes **1162** are adjacent to the filter layer **224** and are disposed on the upper surface **122a**. In other words, the two second electrodes **1162** may be disposed on the second region R2 of the light guide structure **220** exposed outside the filter layer **224**, i.e., the second electrodes **1162** are arranged in a non-light-emitting region on both sides of the light guide structure **220**, so that there is no need to sacrifice a light-emitting area.

(39) In some embodiments, the first electrode **1161** is a first-type electrode and is electrically connected to the first-type semiconductor layer **112**, and the two second electrodes **1162** are second-type electrodes and are electrically connected to the second-type semiconductor layer **116**. A material of the first electrode **1161** and the second electrodes **1162** is, for example, metal, but the disclosure is not limited thereto.

(40) In some embodiments, the two second electrodes **1162** are in direct contact with the first light reflection layer **122** of the light guide structure **220** and penetrate through the first light reflection layer **122** to extend to the second-type semiconductor layer **116**.

(41) Referring to FIG. **12**, compared to the light-emitting element **1100**, the first light reflection layer **122** and the second light reflection layer **330** of the embodiment are doped with the P-type material or the N-type material, so that the first electrode **1261** and the second electrode **1262** may directly make ohmic contact with the second light reflection layer **330** and the first light reflection layer **122**, respectively.

(42) FIG. **13A**, FIG. **14A**, FIG. **15A**, and FIG. **16A** are partial schematic cross-sectional views of display panels according to some embodiments of the disclosure. FIG. **13B**, FIG. **14B**, FIG. **15B**, and FIG. **16B** are respectively partial enlarged view of a region A of FIG. **13A**, a region B of FIG. **14A**, a region C of FIG. **15A**, and a region D of FIG. **16A**.

(43) Referring to FIG. **13A** and FIG. **13B**, in the embodiment, a display panel DP1 includes a substrate S and a plurality of light-emitting elements **10**, where the plurality of light-emitting elements **10** are arranged on the substrate S. It should be noted that although only three light-emitting elements **10** are shown in FIG. **13A**, the disclosure does not limit the number of the light-emitting elements **10**, and the light-emitting elements **10** may be arranged on the substrate S in an array. The substrate S is, for example, a backplane circuit, but the disclosure is not limited thereto.

(44) In the embodiment, the plurality of light-emitting elements **10** have different color light and are electrically connected to the substrate S. For example, the plurality of light-emitting elements **10** respectively have a first color light, a second color light, and a third color light. In here, the disclosure does not limit light colors of the first color light, the second color light, and the third color light. Further, each light-emitting element **10** includes an epitaxial structure **104** and a light guide structure **108**, where the epitaxial structure **104** has a first surface **104a** and a second surface **104b** opposite to each other, and the light guide structure **108** is disposed on the first surface **104a** of the epitaxial structure **104**, where the epitaxial structure **104** may include a first-type semiconductor layer **101**, an active layer **102**, and a second-type semiconductor layer **103** stacked in sequence, and the light guide structure **108** includes a first light reflection layer **105** and a filter layer **106**. The first light reflection layer **105** completely covers the first surface **104a**, and the filter layer **106** covers an upper surface **105a** of the first light reflection layer **105**. Accordingly, in the light-emitting element **10** of the embodiment, the equivalent refractivity of the overall structure is adjusted through a design of the first light reflection layer **105** and the filter layer **106** in the light guide structure **108**, so that the light guide structure **108** only allows photons of a specific wavelength to pass through in each wavelength range. Namely, only light of a specific wavelength may be transmitted out of the light guide structure **120**. In this way, the wavelength (color) of the emitted light may be controlled within a small range, which improves the color purity of the light emitted by the light-emitting element **10**, such that the display panel DP1 including the light-emitting elements **10** has better optical quality.

(45) In some embodiments, the first light reflection layers **105** of the light-emitting elements **10** are connected to each other to form a common light reflection layer. In other words, orthographic projections of the plurality of light-emitting elements **10** on the substrate **S** may fall within a range of orthogonal projections of the first light reflection layers **105** on the substrate **S**, but the disclosure is not limited thereto.

(46) In some embodiments, the display panel **DP1** further includes a protective layer **P** located between the common light reflection layer and the epitaxial structures **104** of the plurality of light-emitting elements **10**. In other words, an orthographic projection of the protective layer **P** on the substrate **S** may be completely overlapped with the orthographic projections of the first light reflection layers **105** on the substrate **S**, but the disclosure is not limited thereto.

(47) In some embodiments, a gap **G** exists between any two adjacent filter layers **106**, and a part of the common light reflection layer **105** is exposed to the outside by the gap **G**. Since the part of the common light reflection layer **105** corresponding to the gap **G** does not have a light transmission property, the light emitted to the gap **G** by each light-emitting element **10** may be reflected by the common light reflection layer and does not penetrate through the first light reflection layer **105**, so as to reduce a chance of lateral light mixing occurred in case of a large viewing angle, but the disclosure is not limited thereto.

(48) In some embodiments, each light-emitting element **10** further includes a second light reflection layer **109**, wherein the second light reflection layer **109** covers the second surface **104b** of the epitaxial structure **104** and the side surface **104s** connecting the first surface **104a** and the second surface **104b**, but the disclosure is not limited thereto.

(49) In some embodiments, the light guide structure **108** extends beyond the side surface **104s** of the epitaxial structure **104**, but the disclosure is not limited thereto.

(50) In some embodiments, the multiple electrodes of each light-emitting element **10** are arranged on the substrate **S** in a horizontal manner, but the disclosure is not limited thereto.

(51) It should be noted that the disclosure is not limited to the various types of light-emitting elements and display panels of the above-mentioned embodiments, and the configurations and components of the above various types of light-emitting elements may be combined according to actual design requirements. In addition, the various types of light-emitting elements may also be designed in the display panel according to actual design requirements, and as long as the epitaxial structure of the light-emitting element of the display panel is designed with a light guide structure including a first light reflection layer and a filter layer, it is considered to be within a protection scope of the disclosure.

(52) Referring to FIG. **14A** and FIG. **14B**, compared to the display panel **DP1**, an area of an orthogonal projection of a filter layer **206** of a light guide structure **208** of a light-emitting element **20** of a display panel **DP2** of the embodiment on the substrate **S** is less than or equal to an area of an orthographic projection of the corresponding epitaxial structure **104** on the substrate **S**. In other words, the filter layer **206** of the light guide structure **208** is retracted into the corresponding epitaxial structure **104**, but the disclosure is not limited thereto.

(53) Referring to FIG. **15A** and FIG. **15B**, compared to the display panel **DP1**, a filter layer of a light guide structure **308** of a light-emitting element **30** of a display panel **DP3** of the embodiment includes a first filter layer **3061** allowing the first color light to pass through, a second filter layer **3062** allowing the second color light to pass through, and a third filter layer **3063** allowing the third color light to pass through, the first filter layer **3061**, the second filter layer **3062**, and the third filter layer **3063** are sequentially stacked on the first light reflection layer **105** to allow a plurality of different color light wavelengths to pass through at the same time. Taking application of a display as an example, the first filter layer **3061**, the second filter layer **3062**, and the third filter layer **3063** may respectively correspond to light with wavelengths of red, blue, and green. In this way, all of the filter layers may be altogether fabricated on the display panel **DP3** to avoid a trouble of setting filter layers of corresponding wavelengths according to the color light of each epitaxial structure

104.

(54) Referring to FIG. 16A and FIG. 16B, compared to the display panel DP3, an area of an orthogonal projection of filter layers (a first filter layer 4061, a second filter layer 4062, and a third filter layer 4063) of a light guide structure 408 of a light-emitting element 40 of a display panel DP4 of the embodiment on the substrate S is less than or equal to an area of an orthographic projection of the corresponding epitaxial structure 104 on the substrate S, but the disclosure is not limited thereto.

(55) In summary, in the light-emitting element of the disclosure, through combination of the first light reflection layer and the filter layer, a forbidden band and a constructive interference section are respectively formed on a light-emitting surface of the epitaxial structure, and only light with a wavelength within such section is allowed to pass through. In this way, the wavelength (color) of the emitted light may be controlled within a small range, thereby improving the color purity of the light emitted by the light-emitting elements, so that the display panel including the light-emitting elements has better optical quality.

(56) It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided they fall within the scope of the following claims and their equivalents.

Claims

1. A light-emitting element, comprising: an epitaxial structure, having a first surface and a second surface opposite to each other, wherein the epitaxial structure comprises an active layer; and a light guide structure, disposed on the first surface of the epitaxial structure, wherein the light guide structure comprises a first light reflection layer and a filter layer covering an upper surface of the first light reflection layer, and the first light reflection layer at least partially covers the first surface, wherein the first light reflection layer forms a forbidden band that shields the light from the active layer, and a constructive interference section, corresponding to a specific transmission spectrum within parts of the active layer and allowing light from the active layer to pass through, is formed on the upper surface covered by the filter layer, and the filter layer retracts within the first light reflection layer, and the upper surface of the first light reflection layer comprises a first region covered by the filter layer and a second region exposed outside the filter layer.
2. The light-emitting element as claimed in claim 1, wherein the filter layer partially covers the upper surface of the first light reflection layer.
3. The light-emitting element as claimed in claim 2, wherein a ratio of an area of the first region to an area of the upper surface is greater than or equal to 20%.
4. The light-emitting element as claimed in claim 2, further comprising at least one electrode electrically connected to the epitaxial structure, wherein the at least one electrode is adjacent to the filter layer and disposed on the upper surface.
5. The light-emitting element as claimed in claim 1, further comprising a second light reflection layer covering the second surface.
6. The light-emitting element as claimed in claim 5, wherein the epitaxial structure further has a side surface connecting the first surface and the second surface, and the second light reflection layer further covers the side surface.
7. The light-emitting element as claimed in claim 6, wherein the first light reflection layer extends beyond the side surface of the epitaxial structure, and the second light reflection layer is connected to the first light reflection layer.
8. The light-emitting element as claimed in claim 1, further comprising a first protective layer, wherein the light guide structure is located between the first protective layer and the epitaxial structure, or the first protective layer is located between the first light reflection layer and the

epitaxial structure.

9. The light-emitting element as claimed in claim 8, further comprising a second protective layer disposed at both sides of the filter layer and covering the first light reflection layer.

10. The light-emitting element as claimed in claim 1, further comprising a plurality of electrodes electrically connected to the epitaxial structure, wherein the plurality of electrodes are disposed at the second surface of the epitaxial structure, or the plurality of electrodes are respectively disposed at the first surface and the second surface of the epitaxial structure.

11. The light-emitting element as claimed in claim 1, wherein a thickness of the filter layer is positively correlated with an equivalent refractivity of the light guide structure.

12. The light-emitting element as claimed in claim 1, wherein the first light reflection layer is doped with a P-type material or an N-type material.

13. The light-emitting element as claimed in claim 1, wherein the first light reflection layer and the filter layer comprise an aluminum material, and an aluminum content of the first light reflection layer is less than an aluminum content of the filter layer.

14. A display panel, comprising: a substrate; and a plurality of light-emitting elements, arranged on the substrate, wherein the light-emitting elements have different color light, and are electrically connected to the substrate, and each of the light-emitting elements comprises: an epitaxial structure, having a first surface and a second surface opposite to each other, and comprising an active layer; a light guide structure, disposed on the first surface of the epitaxial structure, and comprising a first light reflection layer and a filter layer covering an upper surface of the first light reflection layer, wherein the first light reflection layer at least partially covers the first surface, wherein the first light reflection layer forms a forbidden band that shields the light from the active layer, and a constructive interference section, corresponding to a specific transmission spectrum within parts of the active layer and allowing light from the active layer to pass through, is formed on the upper surface covered by the filter layer, and the filter layer retracts within the first light reflection layer, and the upper surface of the first light reflection layer comprises a first region covered by the filter layer and a second region exposed outside the filter layer.

15. The display panel as claimed in claim 14, wherein an area of an orthographic projection of the filter layer of each of the light-emitting elements on the substrate is less than or equal to an area of an orthographic projection of the corresponding epitaxial structure on the substrate.

16. The display panel as claimed in claim 14, wherein the plurality of first light reflection layers of the plurality of light-emitting elements are connected to each other.

17. The display panel as claimed in claim 16, further comprising a protective layer, wherein the protective layer is located between the plurality of first light reflection layers and the plurality of epitaxial structures.

18. The display panel as claimed in claim 16, wherein a gap exists between any two adjacent filter layers, and the plurality of first light reflection layers are exposed to the outside by the gap.

19. The display panel as claimed in claim 14, wherein the plurality of light-emitting elements respectively have first color light, second color light, and third color light, each of the filter layers of the light-emitting elements comprises a first filter layer allowing the first color light to pass through, a second filter layer allowing the second color light to pass through, and a third filter layer allowing the third color light to pass through, and the first filter layer, the second filter layer and the third filter layer are sequentially stacked on the first light reflection layer.

20. The display panel as claimed in claim 14, wherein the epitaxial structure further has a side surface connecting the first surface and the second surface, each of the light-emitting elements further comprises a second light reflection layer, and the second light reflection layer covers the second surface and the side surface.
